

K-Pop Agenda Dynamics

Time Series Analysis of Topic Diffusion and Sentiment

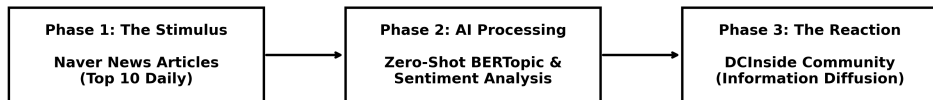
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The Core Question

- **Question:** How do different types of K-Pop news diffuse across anonymous online communities?
- **The Headline Finding:** Controversies are “fast-burning” (peaking in ~ 2.4 days), while professional updates are “slow-building” (peaking in ~ 5.6 days).



Two-Part Dataset:

- **The Stimulus:** Top 10 Naver Entertainment articles daily (Sep '25 – Feb '26).
 - *Note: A filtered subset ($N \sim 15$) accounting for platform search limits and spam was used for the community analysis.*
- **The Reaction:** $\sim 13,500$ DCInside community posts (Sep '25 – Apr '26).

Methods Used:

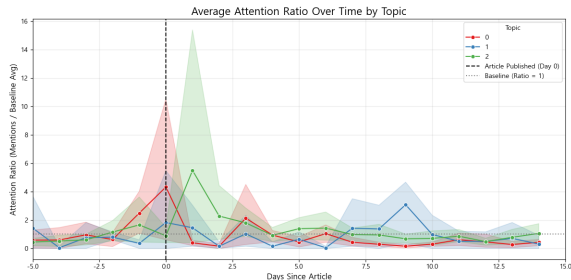
- **Hybrid Topic Modeling:** Zero-Shot BERTopic with Sentence-BERT and Okt Tokenizer to cluster agendas based on cosine similarity.
- **Sentiment Analysis:** HuggingFace NLP (bert-base-multilingual) measured community emotional shift (Before vs. After news).

Finding 1: Diffusion Speed

Fast-Burning vs. Slow-Building

- **Controversial Agendas:** Peaks at 2.40 days.
- **Relational Agendas:** Peaks at 2.71 days.
- **Professional Agendas:** Peaks at 5.67 days.

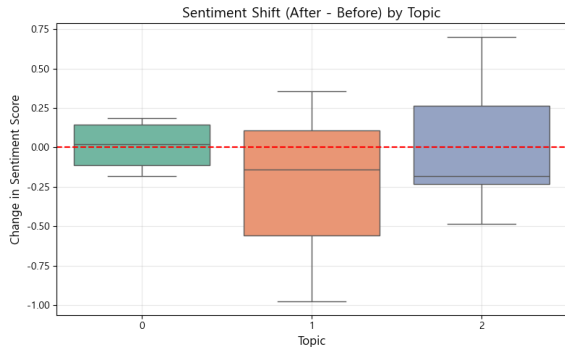
Why? Negativity Bias drives rapid sharing for controversies, while promotional news digests slowly.



Finding 2: Community Sentiment Shift

Volatility vs. Cynicism

- **Professional Agendas:** Uniformly negative shift (-0.25). Shows DCInside's cynical response to corporate PR.
- **Controversial & Relational Agendas:** High volatility. Context-dependent (hero vs. villain), leading to massive swings rather than uniform negativity.



Conclusion for PR & Crisis Management

- **Crisis Window:** Agencies have < 48 hours to respond to a controversy before the community narrative fully peaks and solidifies.
- **Promotional Strategy:** Professional news requires a week-long drip-feed strategy rather than relying on a single press release.
- **Future Work:** Expanding dataset size and comparing DCInside's cynical baseline against platforms like Twitter.